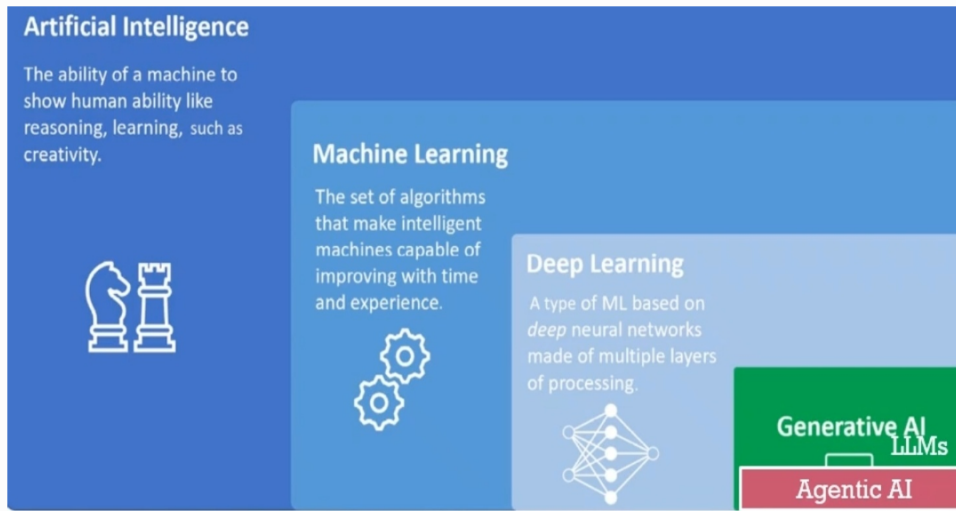


Let's first clarify the hierarchy within artificial intelligence. It begins with a broad field of AI and narrows down to the specific sub fields like deep learning, generative, and agentic AI. They're highlighted as a new top layer of the latest software application of deep learning.



We consider today AI is the foundation while machine learning, deep learning, and generative AI represents specialized and focused area of research and applications. As you can see on this slide,

AI represents the broad field of creating systems which can perform tasks that can show human intelligence. (so is AI the theoretical concept and then machine learning is the actual coding/application that puts AI into action?)

Machine learning is a branch that focuses on creating algorithms and models that enable those systems to learn and to improve themselves with time and training. Machine learning models learn from existing data and then automatically update their parameters as they grow during deployment.

Then deep learning is sub branch of machine learning in a sense that it's a new approach on learning representations from data. It puts an emphasis on learning successive layers of increasingly meaningful representation. The deep stands for the idea of successive layers of representations. Those deep models are called neural networks and how many layers contribute to the model of the data, this is called the depth of the model. Deep neural networks are particularly suitable in the fields like computer vision, language processing, and we will learn this in the course. When we talk about machine learning and deep learning models, we typically refer to the discriminative models whose goal is to make predictions or to inference patterns on top of the data. (what is meant by the term discriminative models? are discriminative models the same as deep learning models? Is it another word for it?)

The generative AI field of research, this is the sub branch of deep learning, and it does not use neural network to cluster or to classify or to make predictions on existing data. But it is using powerful large language models, an example like the transformer neural network models, to generate brand new content from images to text and from speech to transcribe it into text, from music to videos. (so are the LLMs used to make the

predictions or does generative AI only transcribe text?)

The new game changer is Agentic AI which is a class of artificial intelligence that focuses on autonomous agent systems. The agent is designed to make decisions and to perform tasks without human intervention. It can evaluate the goal definition and to determine if the goal is complete. If not, then the agent can change the steps and to new tasks and iterations with large language models. This new world of AI can learn to accomplish any task a human can, we believe today. -understand

Today world runs on data. But this is raw data and it is useless without an insight. Machine learning helps to convert data into actionable intelligence.

Here is a real world example.

1. Banks, they use machine learning today to detect fraud and
2. retailers use it to understand buying behavior.
3. Machine learning also is automating tasks that are repetitive or time consuming and they can be handled by machines. For example, think of the smart spam filters, facial recognition, or robotic process automation in the customer service.
4. Machine learning improves efficiency while it is reducing human error. It has predictive power to forecast sales, user turnover, market trends, and so on. Companies today rely on the predictive models to plan inventory pricing and marketing.
5. In healthcare, machine learning today predicts disease progression or patient risk.
6. Machine learning is the backbone of the modern AI systems recommendation engines on example, Netflix and Amazon. Voice assistance like Siri and autonomous vehicles. It is essential to understanding and building the technologies which define the Ford Industrial Revolution today.

Auto GPT was the first demonstration of the power of using task planning and iterations with LLM models. From this agentic AI exploded into the community using planning, iterations, and task repetition systems. Last year and this year, 2025, they're marked by the development and deployment of autonomous AI agents. (Is auto GPT generative or agentic AI?)

Decision making process is the capability of autonomous agent systems because they can also use other tools to perform different tasks if it's necessary. They can change their algorithm based on new knowledge and human feedback.

Examples of Generative AI

1. There have been significant strides in the generative AI during this year, especially in the creation of the high quality text, images and videos and generative AI tools are being leveraged in marketing, training with companies like Synthesia providing tools for creating deep fake avatars for various applications. This is often used by corporations for communication, orientation and training videos.
2. In advertising campaigns, product demonstrations, and to create chat bots. Synthesia software algorithm mimics speech and facial movements based on the video

recordings of an individual talking and pronunciation. From this, a text to speech video is created to look and to sound like that individual. The user create content by creating digital representations of themselves or personal avatars. These avatars can be used to narrate videos generated from text. Multimodal models like Google D Pines, Gemini, 1.5 Pro are pushing the boundaries of what is possible when you integrate text, images, and other data types into a single system. These models are being used to enhance search experiences, to automate complex queries, and to improve productivity tools.

3. Inspired by all these advancements in generative AI, In the robotics, they're developing more versatile robots capable of multitasking. There is also the growing focus on the responsible development and deployment of the AI system, and the efforts are being made to improve transparency, security, fairness in AI to mitigate risks and to ensure an ethical use.

Automation is deployment of intelligence software and hardware that performs tasks on behalf of humans, like chat bots, self driving cars, robots, and so on. (does automation exist in all levels of machine learning or only agentic AI?) -I personally have a hunch that automation applies to all levels of machine learning and that agentic AI performs tasks through making it's own decisions without the need of any human intervention with finetuning or perfecting the model

Predominantly, we can look at machine learning as set of tools to achieve business insights. Primarily, there are two subtypes,

1. unsupervised, when system seeks to discover relationships and groupings of data and supervised machine learning. The system performs quantification to predict real numbers.

The expected output is the training knowledge being imparted on the system. During training, model continuously measures prediction error and is using the error to continually adjust itself internally.

Training on data is the heart of machine learning. Machine learning algorithms adjusted internal parameters to produce statistically derived outputs. Then we have a learned model which can be put into production systems and utilized repeatedly when new data arrives.

Most common machine learning tasks belong to categories. (which of these categories/ examples fall under the generative AI examples that were listed before?) - I think 1 & 2 represent deep learning since a prediction of a specific value is being made. 3 is an example of generative AI since something is being converted to text

1. Image recognition, we're recognizing different items,
2. regression, we're predicting numerical value given some input, for example, stock price based on the historical data.
3. We can translate from one language to another speech recognition, and this is very important technology today for the voice user interfaces, an example for H Alexa, text to speech. This is another component used today in the communication with machines.

In neural networks, we are going to use these two main concepts, inference and training.

1. Inference is the action of running a neural network on an input, and it can be extremely computationally intensive operation, especially for images.
2. Training is running inference repeatedly and adjusting it based on the error after each time. Most machine learning tasks will be inference of the pre trained neural network models. -understand

Each time we run a neural network on an image as input, we call this inference. (So when we run a neural network on a input that isn't a image then would that be called inference as well or will a different term be used since the input is not a image?)

For very poorly trained neural networks, when you give an image of a dog, it sometimes will say a cat, it's a cat. If neural network is trained, it may be able to identify dog properly. Once you train the network, then millions of people can use it for inference. (so the term inference is not just limited to setting up a model and training it. Even the action of users adding their input data on the model would be considered as inference since they are essentially adding a input in the model? Now when users engage in long chats with machine learning algorithms such as chat GPT, would each chat/follow-up response that the user sends to the machine learning chatbot be considered as inference?)

The breakthrough in speech recognition happened actually before the breakthrough in image recognition. It requires the level of natural language understanding and logical reasoning, which is very different from image recognition.

An acoustic model converts a representation of the sound wave into a probability distribution over fragments of sounds or phonemes. (acoustic models do not fall under the category of generative AI right?) -I personally think that acoustic modeling does fall under the category of generative AI because generative AI converts things to text

With sufficient processing power, neural networks were competitive with the state of the art for acoustic modeling. In 2009, two graduate students showed that pretrained deep neural network could slightly outperform the state of the art on the timid speech dataset. This result reignited the interests of several leading speech groups in neural networks. In 2010, essentially the same deep network was shown to be the state of the art for the large vocabulary speech recognition without requiring speaker dependent training. So by 2012, Google had engineered a production version that significantly improved voice search on Android phone. This was an early demonstration of the power of machine learning. Today, transformer neural networks based speech recognition models have produced dramatic improvements that have revolutionized natural language processing, and they are now being used routinely in the industry. (so text to speech software that is used in phones and tablets for user accessibility would be considered as acoustic models right? would this fall under the category of generative AI as well or would it not be since generative AI only converts things to text?)

How is it possible for neural networks to learn the complicated internal representations that are required for difficult tasks such as recognizing objects or understanding language?

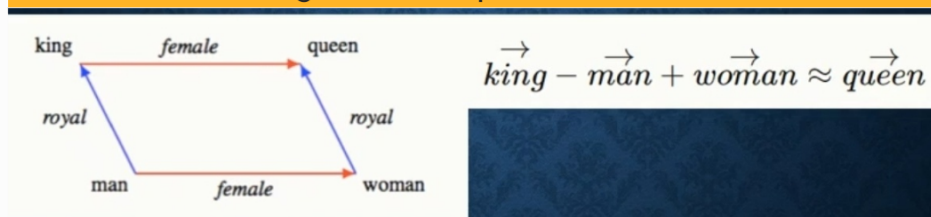
Deep learning seeks to answer this question by using many layers of activity vectors as representations and learning the connection strengths that give rise to these vectors.

Vectors

For the artificial intelligence in the brain inspired paradigm, the external symbols, which are words for speech that are used for communication, are converted into internal vectors of neural activity. (So the vectors actually represent each neuron in a neural network? At what point in the deep learning neural network do the vectors become matrices? or do the vectors, which represent each neuron, always stay as vectors?)

The vectors have a rich similarity structure. Activity vectors can be used to model the structure inherent in a set of symbols by learning appropriate activity vectors for each symbol or word. (what does the instructor mean by "activity vector"? Isn't a vector a vector? or is the instructor referring to the idea that different vectors represent different terms/images? if so, how does one design a vector to represent a particular image/word since vectors just consist of numbers?)

This was first demonstrated by Bengio on real sentences. A very impressive recent demonstration is BERT, which is the cutting edge neural network architecture for natural language processing. The main advantage of using vectors of neural activity to represent concepts and weight matrices to capture relationships between concepts is that this leads to automatic generalization. If king and queen are represented by very similar vectors, they will have very similar causal effects on other vectors of neural activity. (does similar causal effects mean that the machine learning model would confuse the vectors which are similar? How does the Image below represent the causal effects of similar vectors?)



So this facilitates analogical reasoning and suggests that immediate intuitive analogical reasoning is our primary mode of reasoning with logical sequential reasoning being a much later development. (what are the differences between analogical reasoning and logical sequential reasoning? what is analogical reasoning?)

Generative AI is a type of machine learning that can create new content such as text, images, and other media by using generative models. Generative AI models learn the patterns and structure of their input training data and then generate new data that has

similar characteristics. (before this, the instructor talked about acoustic models and how they are models which only convert text to speech. Now the instructor talks about generative AI so is generative AI a type of acoustic model? If yes, then why because from what I understand Generative AI only converts things to text such as images to text, sound to text)

In the early 2020s, the advances in the transformer based deep neural networks, they enabled a number of generative AI systems. (So all generative AI models are based on transformer based deep neural networks? what is the difference between a transformer based deep neural network and a regular deep neural network?)

Which are notable for accepting natural language prompts as input. (ok, I am a bit confused here. So I have heard of the term natural language processing. Is natural language processing strictly limited to generative AI models? And I am guessing that the purpose of natural language processing is to just convert things into text? like identifying what objects are in a image or converting handwriting? or would a LLM do both?)

Examples of LLMs

These include large language models, chat bots, such as Chad GPT, Bart, LAMA (-understand), and then text to image AI art systems, such as stable diffusion, deli, mid journey. (how does text to image art fall under LLMs/generative AI? Since the output isn't text in this scenario?)

Generative AI has uses across a wide range of industries, including art, writing, software development, product design, healthcare, finance, gaming, marketing, fashion, many.

Concerns of generative AI

However, there are also concerns about the potential misuse of the generative AI because of the cyber crime or creating fake news or deep fakes which can be used to deceive or manipulate people. (I get how generative AI can be used to spread misinformation using deepfakes, but how can generative AI be used for cyber crime? are there any real life applications which can illustrate how generative AI is used for cyber crime?)

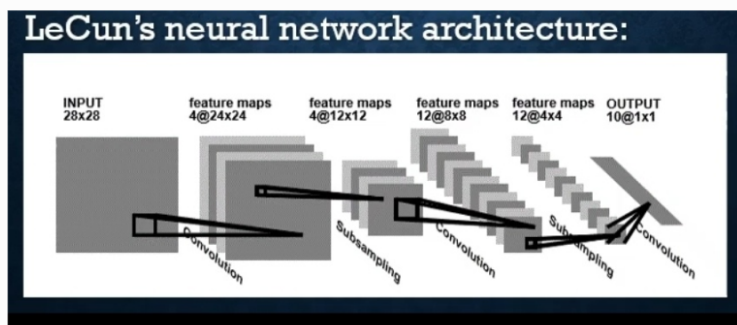
The key to this victory were the major effort in collecting more than 1 million labeled images for the training set and the very efficient use of multiple GPUs by Alex Krzewski. Very successful results have been attained in the area of handwritten digit recognition. At AT&T Bell Labs, a suite of classifier algorithms has been developed thanks to proliferation of powerful inexpensive computers and availability of large databases of characters that can be used for training and testing. Figure on the right is an example of a linear classifier. Each input unit pixel value contributes to a weighted sum for each output unit. The output unit with the largest sum indicates the class of the input digit. The original standard test set contained around 60,000 digit images written by 500 different writers and blocks of data from each writer were scrambled. Linear classifier

achieved over 92% accuracy. The standard methodology had 99%, but performance was much better. Memory requirements and classification speed or recognition time. After that, more sophisticated neural network was developed by Lacun's team that achieved 98.3% accuracy, which is saying even better than the standard methodology. A new measure that is important for system performance has been introduced. It is the time required to train the classifier or training time. This measure is interesting for designers, but it's irrelevant for customers.

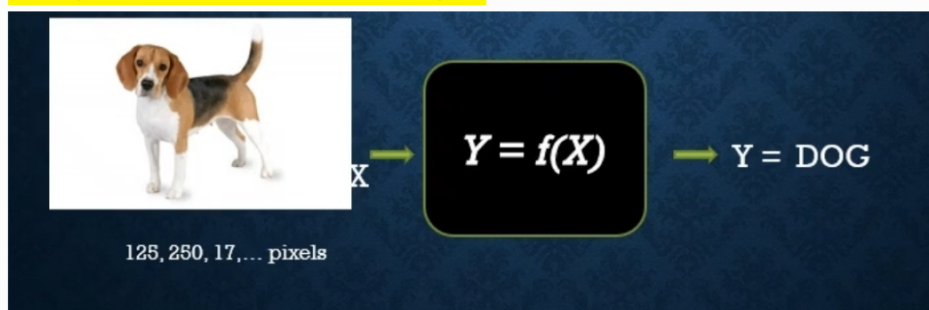
Why is it not possible today to recognize written digits without neural nets?

Well, if there are 28 by 28 binary images, how many possible variations are there to recognize a single digit? There are 784 pixels with two possible settings per pixel, so the maximum number of possibilities is two to the power of 784. There are more variations than their number of particles in universe.

Lacuns neural network architecture is a much simpler solution and does the job with the best accuracy today. (How does Lacuns neural network architecture work? Does it just select one pixel?)



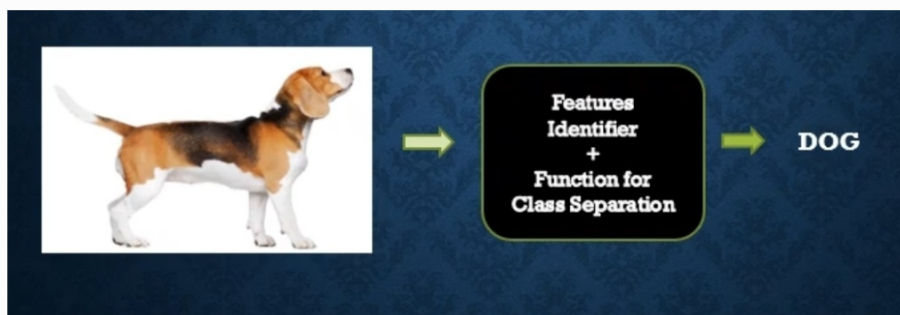
Now, how do we set up the framework for the deep learning? We are essentially writing a computer program that has tens of millions of constants in it. They control its behavior to a certain degree. Each setting of these constants is a different program. Then how we set those constants determines what our program will do. Will it recognize that this is a dog? What is the structure of our computer program whose constants we don't know? First, we're giving it some inputs expressed in our numerical representation. Then how do we calculate something about them? That is this a dog? Our computer program uses a mathematical function, Y equals F of X , where X is the representation of our input, the picture data, and Y is representation of our output.



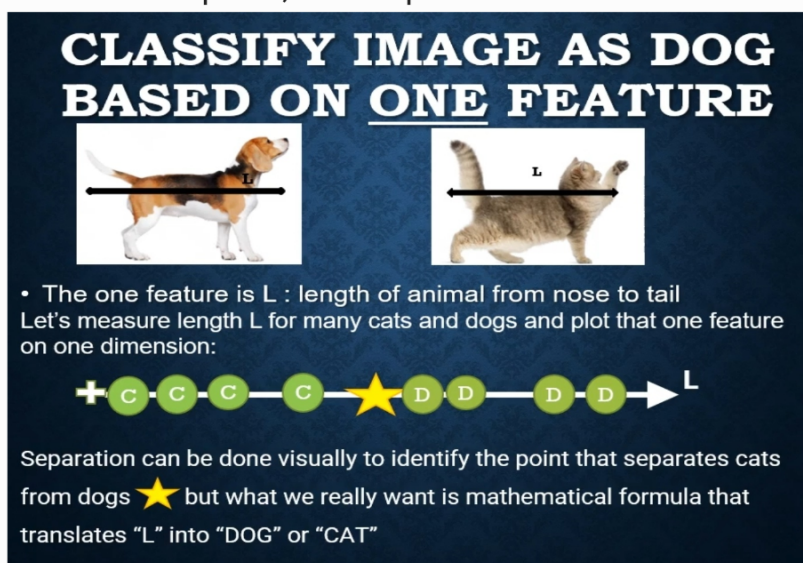
For example, in the word dog. How do we select the function F of X ? Function F has two distinguished parts.

First part identifies features and the features are sent to another block called function for class separation that based on the feature makes a decision if it is a cat or a dog. We are going to start with a very simple class separation function that takes only one feature. (Is the class separation function the same as a regression line?)

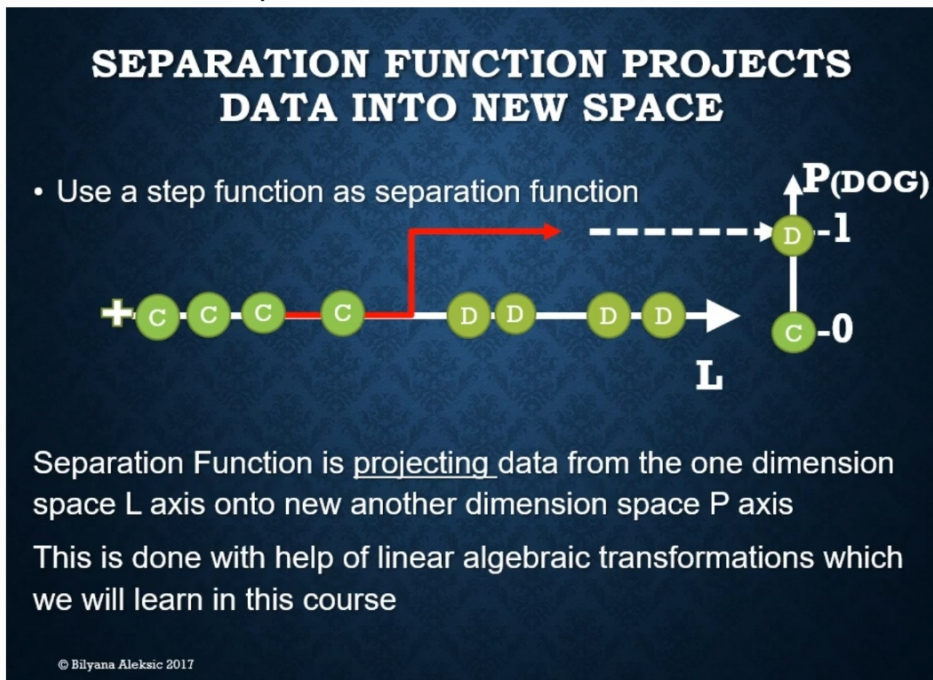
Let's pick that feature to be length of the animal. It's the only feature machine will decide upon whether it's a cat or a dog or an elephant.



Since we decided to use length of an animal as the only one feature we are going to measure, we take the animal and measure its length and classify animals based on that one feature. We took a bunch of cats and dogs and measured their lengths and plotted them on one dimensional line L , which is the length. For the group of animals we had, it turns out that all the cats had shorter lengths than all the dogs and that dog samples are more to the right of the L -axis. The circles in blue, so for bigger values of L , you can see on the slide that there is a separation point marked with the golden star. If animal has length less than the length of that star point, at this point it's a cat and for the L with greater values, it's a dog.



This is very nice, but what we really want is a formula, a function that will tell us what the data is saying (so will the function plot multiple golden stars for all the data points?). That function is a step function.



When I apply this function to the data, I will get a new axis, which is the probability that the data is a cat or a dog. It's axis P. If we position the step function properly, which is putting it right between the green and the blue points (the points are are classified as dog) on the L axis, I can multiply green points by zero and blue points by one, and I can get the new axis P (where is the star in this image to mark the separation point?). All the dogs are going to be projected into one point, the blue point with letter D, and all the cats are going to be projected on another point C in green on the axis B. The separation function is a nonlinear function here and we will use these a lot in machine learning. But the point of this slide is to show that by projecting data from one axis, the one dimensional space L into a brand new axis, which is the probability of the different classes P will tell me which class the input data image belongs to. Projecting from one space to another space is crucial to everything in machine learning. We learn matrix transformations which enable these in this course.

For the group of cats and dogs in the previous slide that we measured, we had a very good separation and tight clusters of dogs and cats on the P axis. Our neural network model seemed very good. But then another customer called who has a different set of cats and dogs and among them there are small dogs. They will cause the problem to our model. Now there are some dogs that are similar in size to some cats, and the green and the blue dots will become intermingled on the L axes. The step function cannot help us here. (why can't the step function help here? will the separation line still be used? I thought the step function creates the red line/separation line?)

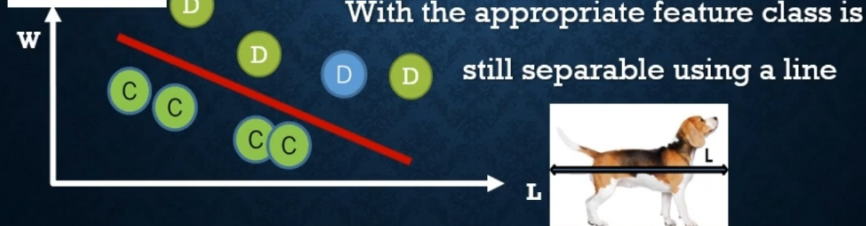
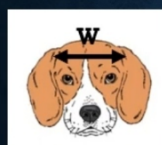
MULTIPLE FEATURES

- Sometimes neither human nor the machine can easily identify the unique class separation point

+ C C D D C D C D → L

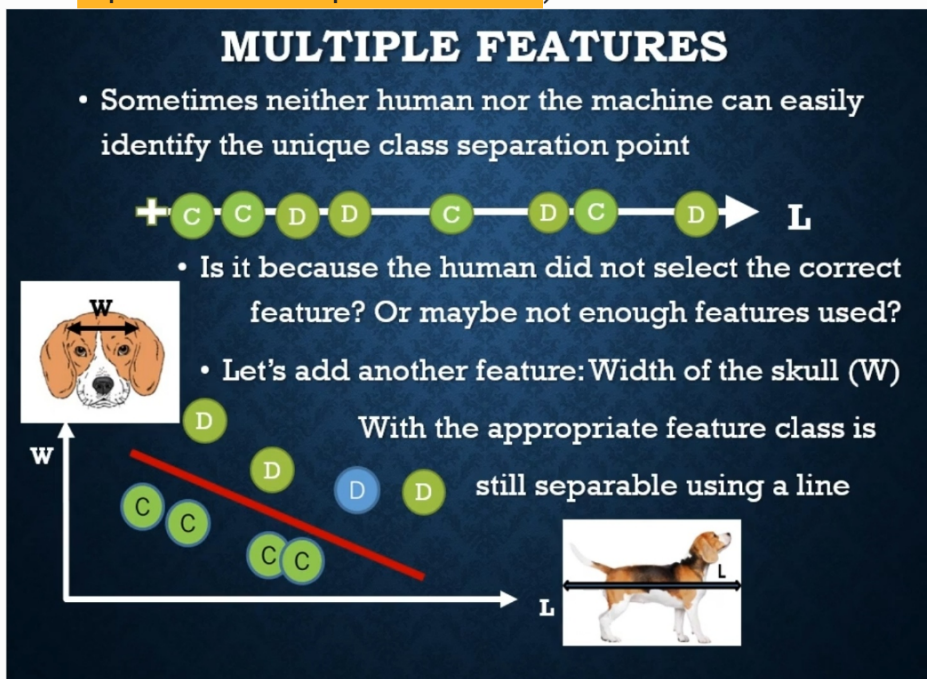
- Is it because the human did not select the correct feature? Or maybe not enough features used?

- Let's add another feature: Width of the skull (W)



To really separate the two groups, we need a nonlinear function that we can fit during training. But let's stay with linear approximation for now. The first thing we will try is a line with the slope, instead of a step where slope is controlled by how big is the mix of the dogs and cats in the separation area (since we are working with linear approximation now will it be correct to say that we are using a separation line? can a separation line also be nonlinear? What is the difference between a line with a slope and a step function? aren't a line with a slope and step functions both considered separation lines?). If it's a wider mix, then slope will be less. If it's a small mix, the slope will be more abrupt and becoming eventually a step function if the separation is clean. The slope of the red line will control moving of the two clusters on the P axis now. How do we move the separation line and change the slope? We need to introduce another axis W. Our separation function is now two dimensional function because the straight line has two parameters, the slope and the intercept of that line with the L axis (when you are saying that the separation line is two dimensional, are you referring to the separation line in the image?).

Now, we are historically at 1960s where perceptron was introduced in machine learning. It had an input layer, which is the length of the animal and it had the output layer where the actually determined the class. Then maybe length L is not the right feature, or maybe one feature is not enough. Now I have two parameters. This looks like a very nice plot in the left bottom corner of this slide, a straight line in the two dimensional space. When I pick appropriate features, I can separate the green from blue dots. But how does this new feature W fit into my perceptron model? Machine learning is the process of finding the appropriate straight red line to fit the data you're given. If classes are truly linearly separable, if we can draw a line of separation, we can come up with a function that projects the clusters of data into a new axis, and let's call it Z and projections for different classes will be away from each other, the blue and the green circles. We can control how the projected clusters will move on the Z axis by changing our model, the slope of the red line in our two dimensional system. You can recognize the formula for Z on this slide, this linear relationship. Z is a plane and you can play with the two parameters W one and W two, and we can call these weights that multiply the width W of the head and the length of the animal L (does the formula of Z represent the separation line?).



When we find the optimal values for these weights, we can project this set of data into two clusters that are clearly separated. This is the main problem in machine learning to find weights and biases (Do weights and biases have a impact or effect on the separation line? Is the separation line the same as the line of regression?). The equation of the plane is equation of the two lines that intersect (what two lines of intersection are being referred to here?). Practically, we take feature L , multiply it by W two, and we take feature capital W and multiply it with the W one and we get our output Z , and this is the model for two features. You get a line. But in reality, we have multiple features. The problem of finding the weights for such systems, we will learn in this course, and it's called gradient descent and we'll talk about it in the Week seven of the course.